Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 12 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 12 : SLO â€“ 2 | DEQUE â€” Insertion Operations

■ I. Insert at Right of Double-Ended Queue

• void insert_right() {
• int val;
• printf("\n Enter the value to be added:");
• scanf("%d", &val);
• if ((left == 0 && right == MAX-1) || (left == right + 1)) {
• printf("\n OVERFLOW");
• return;
• }
• if (left == -1) {
• left = 0;
• right = 0;
• }
• else {
• if (right == MAX-1)
• right = 0;
• else
• right = right + 1;
• }
• deque[right] = val;
• }

■ II. Insert at Left of Double-Ended Queue

• void insert_left() {
• int val;
• printf("\n Enter the value to be added:");
• scanf("%d", &val);
• if ((left == 0 && right == MAX-1) || (left == right + 1)) {
• printf("\n Overflow");
• return;
• }
• if (left == -1) {
• left = 0;
• right = 0;
• }
• else {
• if (left == 0)

• left = MAX-1;
• else
• left = left - 1;
• }
• deque[left] = val;
• }